

⓪ $\text{state} = x, y, \phi, v_x, v_y, \omega$

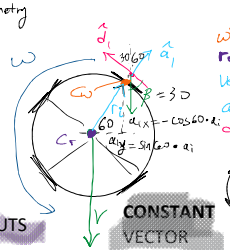
williams_model.py: calculates:
- wheel contact velocity, - d and a wheel slip velocities
- Wheel force from wheel slip velocities, friction coefficients assumed

① One omni-wheel contact
a) robot velocity = $V$
b) robot body angular velocity = $\omega$

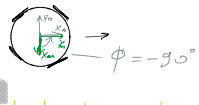
$R_w$ = wheel radius = $\frac{59.8292}{2} = 29.9146\text{mm}$

$R_r$ = roller radius = $\frac{14378}{2} = 7.188\overline{8}\text{mm}$

② Geometry




wheel center = $c_w$ =
Robot center = $c_r$
Vector of wheel = $r_i$
axial direction = $\hat{a}$
drive direction = $\hat{d}$



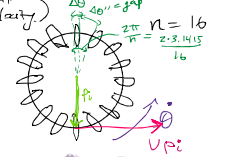
$\phi = -90°$

③ Velocity of Wheel Center (angular velocity)

wheel radius vector = $P_i$

$V_{center_i} = V + \omega \times r_i$

$= \dot{\theta}_i$




④ Contact point velocity

$V_{contact_i} = V_{center_i} + V_{p_i}$

wheel angular velocity = $\dot{\theta}_i$

Peripheral wheel speed = $V_{p_i} = \dot{\theta}_i \times P_i$

⑤ Projection of contact velocity onto axes

$V_{slip,d} = V_{contact_i} \cdot \hat{d}_i$ — $V_{p_i}$ is same axis as $\hat{d}_i$ ∴ dot product with itself is ①!

$= \left(V + (\omega \times r_i) + (\dot{\theta}_i \times P_i)\right) \cdot \hat{d}_i$ — $\hat{d} \cdot \hat{d} = 1$

$= V \cdot \hat{d}_i + (\omega \times r_i) \cdot \hat{d}_i + \dot{\theta}_i P_i$ — scalar

$V_{slip,a} = V_{contact_i} \cdot \hat{a}_i$

$= \left(V + (\omega \times r_i) + (\dot{\theta}_i \times P_i)\right) \cdot \hat{a}_i$ — $\hat{d}$ direction

$= V \cdot \hat{a}_i + (\omega \times r_i) \cdot \hat{a}_i + (\dot{\theta}_i \times P_i) \cdot \hat{a}_i$

So the slip velocities (drive and axial directions) are calculated as follows.

$V_{slip,d} = V \cdot \hat{d}_i + (\omega \times r_i) \cdot \hat{d}_i + \dot{\theta}_i P_i$ — Scalar

These radius

$\hat{d}_i = \hat{x}\sin 30° + \hat{y}\cos 30°$

$\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3, \dot{\theta}_4 \rightarrow V$

Desired Velocity in body frame

$S = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$

Determine Wheel center velocity (ie, the wheel!)

$V_{center_i} = V + \omega \times r_i = \begin{bmatrix} V_x - r_{iy}\omega \\ V_y + r_{ix}\omega \end{bmatrix}$ $\begin{bmatrix} -\omega r_y \\ \omega r_x \end{bmatrix}$

$\omega_i = V_{center_i} \cdot \hat{d} = V \cdot \hat{d}_i + (\omega \times r_i) \cdot \hat{d}_i$

then express this as a matrix

$= \begin{bmatrix} V_x \\ V_y \end{bmatrix} \cdot \begin{bmatrix} \hat{d}_x \\ \hat{d}_y \end{bmatrix} + \begin{bmatrix} -r_{iy}\omega \\ r_{ix}\omega \end{bmatrix} \cdot \begin{bmatrix} \hat{d}_x \\ \hat{d}_y \end{bmatrix}$

$= \hat{d}_x (V_x - r_{iy}\omega) + \hat{d}_y (V_y + r_{ix}\omega)$

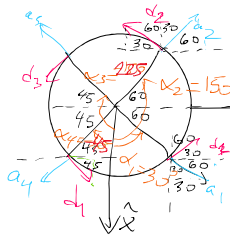
$= V_x \cdot \hat{d}_x + \hat{d}_x(-r_{iy}\omega) + \hat{d}_y V_y + \hat{d}_y r_{ix}\omega$

$= V_x \cdot \hat{d}_x + V_y \hat{d}_y + \omega \cdot (r_{ix}\hat{d}_y - r_{iy}\hat{d}_x)$

$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \end{bmatrix} = \begin{bmatrix} \hat{d}_{x_1} & \hat{d}_{y_1} & (r_{1x}\hat{d}_{y_1} - r_{1y}\hat{d}_{x_1}) \\ \hat{d}_{x_2} & \hat{d}_{y_2} & (r_{2x}\hat{d}_{y_2} - r_{2y}\hat{d}_{x_2}) \\ \hat{d}_{x_3} & \hat{d}_{y_3} & (r_{3x}\hat{d}_{y_3} - r_{3y}\hat{d}_{x_3}) \\ \hat{d}_{x_4} & \hat{d}_{y_4} & (r_{4x}\hat{d}_{y_4} - r_{4y}\hat{d}_{x_4}) \end{bmatrix} \begin{bmatrix} V_x \\ V_y \\ \omega \end{bmatrix}$

$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \end{bmatrix} = \begin{bmatrix} \hat{d}_{x_1} & \hat{d}_{y_1} & r \\ \hat{d}_{x_2} & \hat{d}_{y_2} & r \\ \hat{d}_{x_3} & \hat{d}_{y_3} & r \\ \hat{d}_{x_4} & \hat{d}_{y_4} & r \end{bmatrix} \begin{bmatrix} V_x \\ V_y \\ \omega \end{bmatrix} \rightarrow M = \left[\right]$

where $r$ is the radius of the "robot", but at the wheel = $79.485\text{mm}$




$\hat{d}_1 = -\hat{x}\cdot\sin 60° + \hat{y}\cdot\cos 60°$

$\hat{d}_2 = -\hat{x}\cdot\cos 60° - \hat{y}\cdot\sin 60$

$\hat{d}_3 = +\hat{x}\cdot\cos 45 - \hat{y}\cos 45$

$\hat{d}_4 = +\hat{x}\cdot\cos 45 + \hat{y}\cos 45$

remove if wrong

$\hat{a}_1 = \ominus\hat{x}\cos 30 \oplus \hat{y}$ (+)

$\hat{a}_2 = \ominus\hat{x}\cos 150 \oplus \hat{y}$ (−) V

(→V

$\hat{a}_3 = -\hat{x}\cos(-150) \oplus \hat{y}$

$\hat{a}_4 = \ominus\hat{x}\cos(-30) \oplus \hat{y}$ (+) Planar

$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \end{bmatrix} = \begin{bmatrix} \hat{d}_1 \\ \hat{d}_2 \\ \hat{d}_3 \\ \hat{d}_4 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix}$

for $\hat{y}$

$(\sin 30° +$

$F_{frk} = ma = 2.698 \cdot 2$

$N_i = \frac{mg}{4} = 2.698$

$F_{d,i} = \mu_{d_{dyn}} N_i$

$F_{a,i} = \mu_{a_{dyn}} N_i$

$F_{total} = \sum \mu_{d_{dyn}} N_i \hat{d}$

$= \mu_{d_{dyn}} N_i \cdot M_j \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix} +$

$\alpha$
sin 30
(+)
sin 150
(+)
(−) ✓
sin(−150)
n(−30)
(−) ✓

if $\alpha < 0$ j $\alpha + 360°$

$$\hat{a}_i = \hat{x}\cos\alpha + \hat{y}\sin\alpha$$

$$\hat{d}_i = \hat{x}\cos(\alpha + 90°) + \hat{y}\sin(\alpha + 90°)$$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$$

$$\left(\sin 150°\right) + \left(-\sin(-150°)\right) + \left(-\sin(-30°)\right)$$

$2\,m/s^2$

$.9.81\,m/s^2$

4

$+ M_{ad_{yn}} N_i \hat{a}\Big) +$

$M_{ad_{yn}} N_i M_a \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix}$

$M_a =$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} \cos\alpha_1 & \sin\alpha_1 \\ \cos\alpha_2 & \sin\alpha_2 \\ \cos\alpha_3 & \sin\alpha_3 \\ \cos\alpha_4 & \sin\alpha_4 \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix}$$

$$\begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix} = \begin{bmatrix} \cos(\alpha_1 + 90°) & \sin(\alpha_1 + 90°) \\ \cos(\alpha_2 + 90°) & \sin(\alpha_2 + 90°) \\ \cos(\alpha_3 + 90°) & \sin(\alpha_3 + 90°) \\ \cos(\alpha_4 + 90°) & \sin(\alpha_4 + 90°) \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix}$$

$M_D =$

So the slip velocities (drive and axial directions) are calculated as follows.

$$V_{slip,d} = V \cdot \hat{d}_i + (\omega \times r_i) \cdot \hat{d}_i + \underbrace{\dot{\theta}_{pi}}_{\text{Scalar}}$$

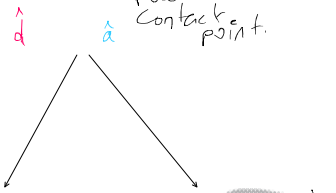
$$V_{slip,a} = V \cdot \hat{a}_i + (\omega \times r_i) \cdot \hat{a}_i$$

These give scalar numbers in the $\hat{d}$, $\hat{a}$ directions.

$$\begin{vmatrix} i & j & k \\ 0 & 0 & \omega \\ a_{ix} & a_{iy} & 0 \end{vmatrix} = i(-a_{iy}\omega) - j(-a_{ix}\omega) = \begin{bmatrix} -\omega r_{iy} \\ \omega r_{ix} \end{bmatrix}$$

$$\begin{bmatrix} \dot{\theta}_3 \\ \dot{\theta}_4 \end{bmatrix} \begin{pmatrix} d_{x3} & d_{y3} \\ d_{x4} & d_{y4} \end{pmatrix} \begin{bmatrix} r & \\ & r \end{bmatrix} \begin{bmatrix} \omega \end{bmatrix}$$

where $r$ is the radius of the "robot", but at the wheel contact point. $= 79.485$ mm




(5) Force generated by each wheel at point $P_i$ under sliding assumption

$$F_i = -\frac{mg}{n_{wheels}}\left(\mu_d(V_{slip,d})\cdot\hat{d} + \mu_a(V_{slip,a})\cdot\hat{a}\right) = -\frac{2.698\,kg \cdot 9.81}{4}\left(\mu_d(V_{slip,d})\begin{bmatrix} a_{ix} \\ a_{iy} \end{bmatrix} + \mu_a(V_{slip,a})\begin{bmatrix} a_{ix} \\ a_{iy} \end{bmatrix}\right)$$

4 wheels

where $\mu_d(V)$, $\mu_a(V)$ are functions of the drive and axial components of friction dependent on velocity.

★ Need to experimentally determine $\mu_d(V)$, $\mu_a(V)$

Williams 2002 approximates $\mu_d(V)$ and $\mu_a(V)$ to the following:

$\alpha$ = is a measure of how fast $\mu_{stat} \to \mu_{dyn}$

$$\mu_d(V_{slip,d}) = \frac{2}{\pi}\tan(KV_{slip,d}) \cdot \left(\mu_{dyn,d} + (\mu_{stat,d} - \mu_{dyn,d})e^{-\alpha_d |V_{slip,d}|}\right)$$

$$\mu_a(V_{slip,a}) = \frac{2}{\pi}\tan(KV_{slip,a}) \left(\mu_{dyn,a} + (\mu_{stat,a} - \mu_{dyn,a})e^{-\alpha_a |V_{slip,a}|}\right)$$

where $\mu_{dyn,d}$, $\mu_{dyn,a}$ } static/dynamic friction for drive/axial directions, and $K$ are constants.

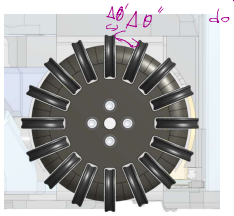
$\mu_{stat,d}$, $\mu_{stat,a}$ } how fast static changes to dynamic friction when a wheel slips.
$\alpha_d$, $\alpha_a$

★ Williams chose $K = 1000$ for numerical stability in Simulink

I found $K = 200$ works better without needing as really small timesteps

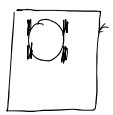
★ Important note. Williams provides an improved friction model however their wheel has half the rollers as ours and the plastic material touches the floor in between.

∴ we do not need the improved friction model as our wheels do not touch the wheel material in between before the next roller.

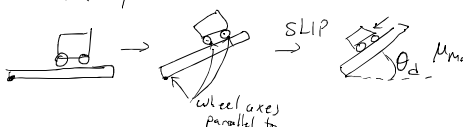



Estimate $\mu_{d,max}$ → all wheels aligned in common direction

ie



rigid board with carpet

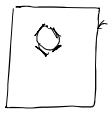



$\mu_{max,d} = \tan\theta_d$

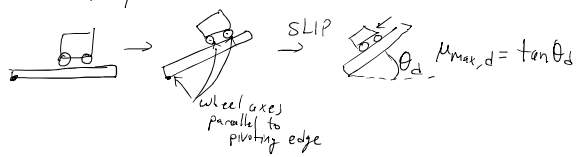
# Estimation from Ri-one - 2023

Estimate $\mu_{d,max}$ → all wheels aligned in common direction

ie

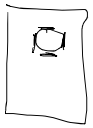


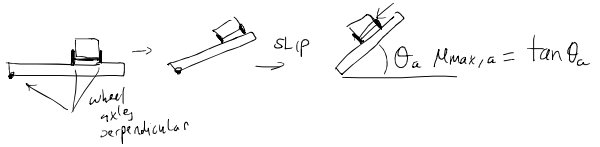
rigid board with carpet




$\mu_{max,d} = \tan\theta_d$

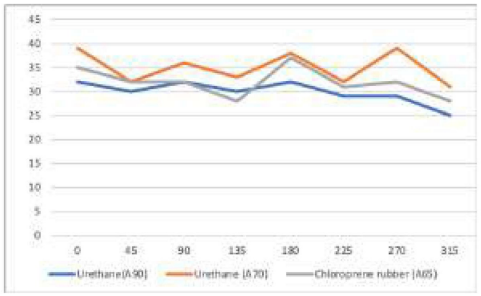
Estimate $\mu_{a,max}$ → all wheels perpendicular to surface






$\mu_{max,a} = \tan\theta_a$

Check wheel slant



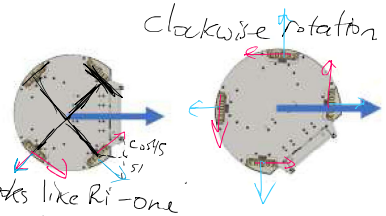

$$4\mu_{eff} = 4\frac{\mu_d}{\sqrt{2}} + 4\frac{\mu_a}{\sqrt{2}} \qquad (0, 90°, 180°, 270°)?$$

($4\mu_{eff}$: known)

We know $\mu_d >> \mu_a$

$$4\mu_{eff} = 2\cdot\mu_d + 2\cdot\mu_a$$

clockwise rotation




kes like Ri - one

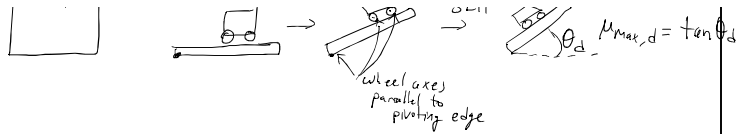
e all 45° deg $\longrightarrow$ $\cos 45 = \sin 45 = \frac{1}{\sqrt{2}}$

**Fig. 12.**

if $|F_i| = \sqrt{F_{x_i}^2 + F_{y_i}^2}$

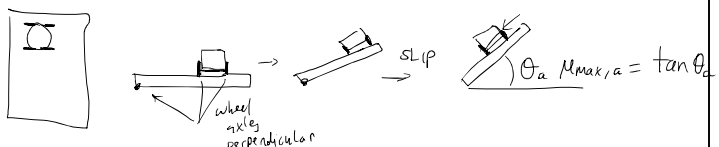
$F_{ri}$

and $F_{ri} =$ F ... $\sqrt{2}$




check google sheets

Estimate $\mu_{a,max}$ → all wheels perpendicular to surface




[Coefficients of Friction + Alpha Estimation from Video Tracking]

(5.5) convert $F_i$ and $r_i$ to world frame

$R_{ot} = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$

robot_dynamics.py:
- Wheel torques
- x,y,phi accelerations

(6) Dynamic Equations (in World frame)

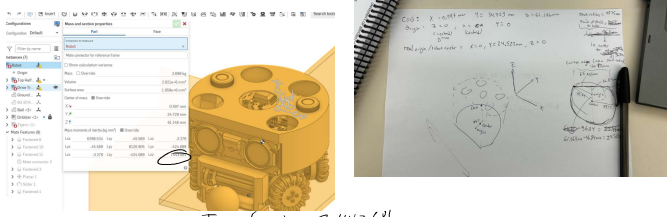
$$m\ddot{x} = \sum_{i=1}^{4} \hat{x} \cdot F_i \qquad m\ddot{y} = \sum_{i=1}^{4} \hat{y} \cdot F_i \qquad I_{zz}\ddot{\phi} = \sum_{i=1}^{4} \hat{z} \cdot (r_i \times F_i)$$

The non linear dynamics equations are of the form $\ddot{X} = f(X, \dot{X})$

Solve the system of equations for $\ddot{x}$, $\ddot{y}$, $\ddot{\phi}$

$m = 2.698\,kg$

$I_{zz}$ in z direction Moment of inertia, related to CoG.



$I_{zz\,(cog)} = 7443.681$

COG = $x = 0.597$, $y = 10.206$, $z = 61.146$.

So $I_{zz}$ (about center) $= I_{zz\,(cog)} + m\,(x_{cog}^2 + y_{cog}^2)$

$= 7443.681 + 2.698\,kg \cdot (0.597^2 + 10.206^2) = 7725.673\ kg\,mm^2 \cdot \left(\frac{m}{1000\,mm}\right)^2 = 0.007725673\ kg\,m^2$

$$\ddot{x} = \frac{\sum_{i=1}^{4} \hat{x} \cdot F_i}{2.698\,kg}$$

input for

To get $\ddot{\theta}$, we need to have

$J_{wi}\ddot{\theta}_i = \tau_{motor,i} - R_{wi}F_{di}$ (axis force on ... radius of wheel)

all 45° deg **Fig. 12.** $\cos 45 = \sin 45 = \frac{1}{\sqrt{2}}$

$F_{xi}$

and $F_{xi} = F_d$, $F_{yi} = 0 \Rightarrow |F_i| = \sqrt{F_{xi}^2} =$

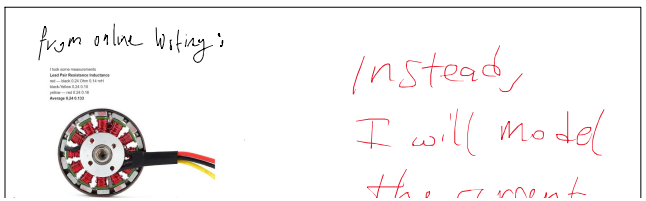
$\therefore$ $F_4 = F_d$ $F_3 = F_d$

$F_2 = F_d$ $F_1 = F_d$ $\therefore$ Summation works for ONLY the 0° case!

Data

wheel slip calculation

$\dot{\theta}$




Instead, I will model the current

$$\ddot{x} = \frac{\sum_{i=1}^{4} \hat{x} \cdot F_i}{2.698\,kg}$$

$$\ddot{y} = \frac{\sum_{i=1}^{4} \hat{y} \cdot F_i}{2.698\,kg}$$

$$\ddot{\phi} = \frac{\sum_{i=1}^{4} \hat{z} \cdot (r_i \times F_i)}{7725.673\,kg} = \dot{\omega}$$

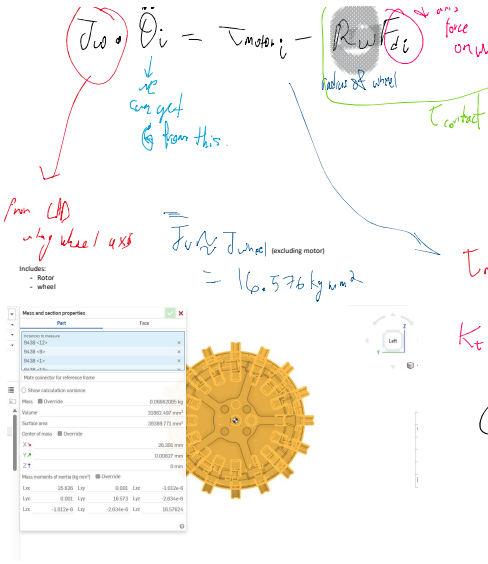
(z component of torque)

State $\dot{X} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{y} \\ \ddot{y} \\ \dot{\phi} \\ \ddot{\phi} \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ \phi \\ V_x \\ V_y \\ \omega \end{bmatrix}$ → need to transform $V_x, V_y$ into body frame for calculations

(7) integrator.py - RK4

$\dot{X} = f(X, u)$

$X_k \rightarrow X_{k+1}$




$$\ddot{\theta}_i = \frac{K_t \cdot I_i - R_w F_{di}}{J_w}$$

(optional) (8) Rolling force (inverse kinematics, no slip)

$$\ddot{\theta}_{i,rolling} = \frac{-(\dot{V} \cdot \hat{d}_i + (\dot{\omega} \times r_i) \cdot \hat{d}_i)}{P_i}$$

→ can we take the derivative and get $\ddot{\theta}_{rolling}$?

$$\frac{d}{dt}(\omega \times r_i) = (\dot{\omega} \times r_i) + (\omega \times (\omega \times r_i))$$

$$\ddot{\theta}_{rolling} = \frac{1}{P_i} \frac{d}{dt}\left(\begin{bmatrix} V_x \\ V_y \end{bmatrix} \cdot \hat{d}_i\right) + \frac{d}{dt}\left(\begin{bmatrix} -r_{iy}\omega \\ +r_{ix}\omega \end{bmatrix} \cdot \hat{d}_i\right)$$

$$= \frac{1}{P_i}\left(\begin{bmatrix} a_x \\ a_y \end{bmatrix} \cdot \hat{d}_i + (\dot{\omega} \times r_i) + (\omega \times (\omega \times r_i))\right)$$

Required $F_d$ contact force when slip is close to zero. ($T_{load}$ is -ve)

$$\ddot{\theta} = \frac{T_{motor} + T_{load}}{J_w} = \frac{K_T \cdot I_q + R_w F_d}{J_w}$$

($F_d$ is negative to motion)

$$\therefore F_{d_{req}} = \frac{\ddot{\theta} J_w - T_{motor}}{R_w}$$



R = 0.245Ω  L = 0.133mH

$$\dot{i} = \frac{V_m - R i - k_t \omega}{L}$$

(ignore this)

I will model the current set point and tune that

motor current

$T_{motor} = K_t \cdot I_i$

$K_t = \frac{60}{2\pi K_v} = \frac{60}{2\pi \cdot 360} = 0.0265 \frac{Nm}{A}$

Control $i_q$ using a PI controller

with $K_p =$

$K_I =$

$J\omega$ $J_\omega$ $-\rho_i \left( \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} \cdot d_i + (\dot{\omega} \times r_i) + (\omega \times (\omega \times r_i)) \right)$

$\therefore F_{d_{required}} = \dfrac{\ddot{\theta} J_\omega - T_{motor}}{R_\omega}$

$F_{d_{required}} = \dfrac{\ddot{\theta}_{rolling} J_\omega - T_{motor}}{R_\omega}$